Longfin squid (*Doryteuthis (Amerigo) pealeii*)

Summary Scores

- **Sensitivity:** Low
- **Exposure:** High
- **Vulnerability:** Low

Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.04	3	
Complexity in Reproductive Strategy	2.44	2.72	
Dispersal of Early Life Stages	1.68	2.4	
Egg and Larval Survival and Settlement Requirements	2	2	
Historical Variability in Temperature	1.48	3	
Juvenile Habitat Requirements	2.04	3	
Other Stressors	2.16	2.35	
Population Growth Rate	2	2.3	
Prey Specificity	1.8	3	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	2.04	3	
Spawning Cycle	1.52	2.7	
Stock Size/Status	1.36	2.9	
Total Sensitivity | Certainty	**Low (1) | 0.98**		

Exposure

Exposure Factors	Mean	SD	Tally
Sea Surface Temperature	2.93	1.12	
Sea Surface Salinity	3.03	0.89	
Surface pH	3.86	0.36	
Mixed layer depth (delta rho = 0.03)	1.11	0.39	
Small zooplankton integrated biomass	1.37	0.41	
Medium zooplankton integrated biomass	1.32	0.38	
Large zooplankton integrated biomass	1.1	0.19	
Total Exposure	**High (3.02)**	**0.54**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Low (3.02)**	**0.54**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Longfin squid. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Low	Low	None
Exposure	High	High	None
Vulnerability	Low	Low	None
Directional Effect	Positive	Neutral	Decrease

Stocks

Longfin squid is managed as a single stock or stock information was not considered for this iteration of the CVA.

Score Descriptions:

Sensitivity:

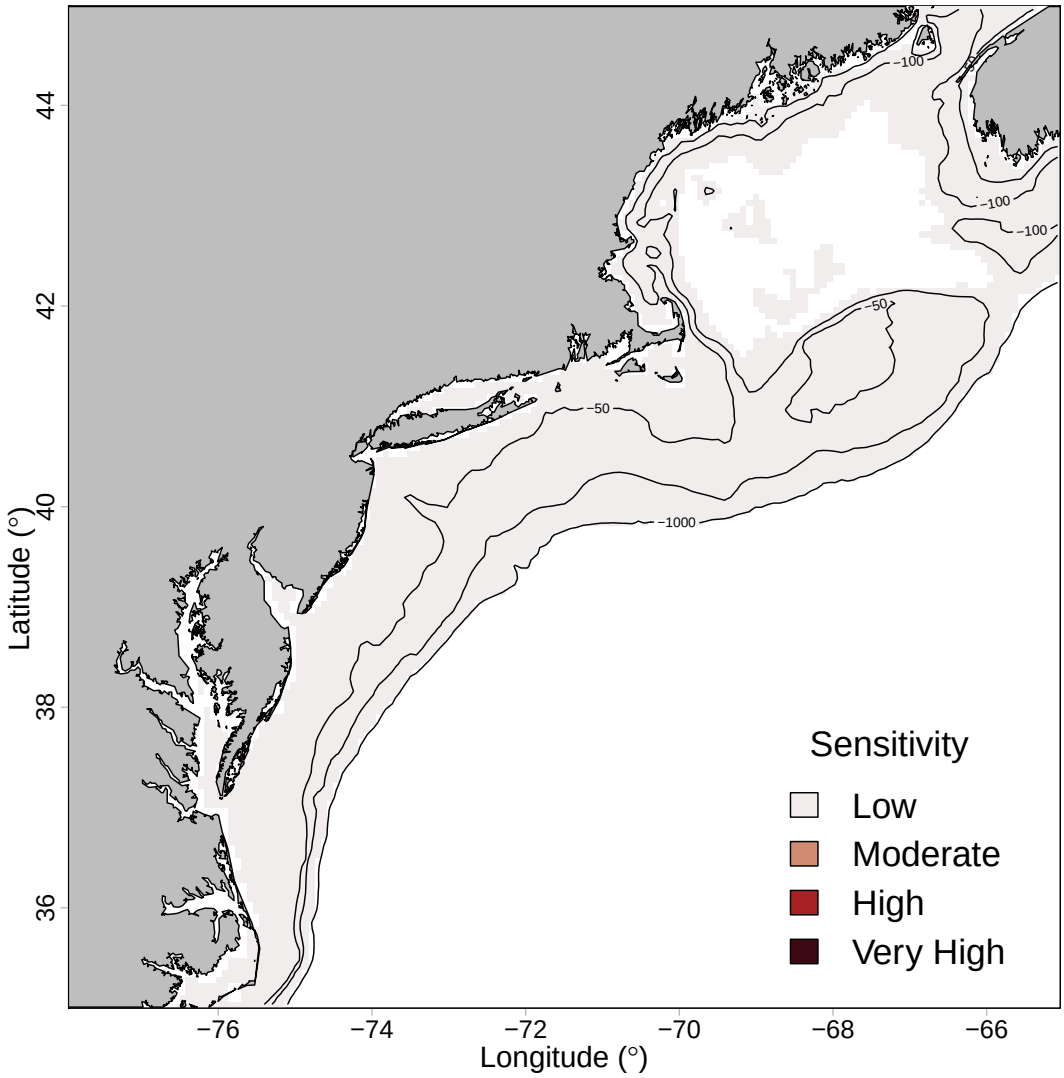


Figure 2: Longfin squid Sensitivity on the Northeast Shelf

Long-finned squid received a **Low** sensitivity score. Certainty was 98% according to bootstrap results. No attributes scored greater than 2.5. Complexity in Reproductive Strategy received the highest score (2.4) (Figure 1). Eggs are cemented to bottom substrate, which can include algae and boulders that may be impacted by dredging activities in nearshore spawning grounds (Jacobson 2005) (Guyant et al. 2022).

Long-finned squid also received a Low sensitivity score in the previous Climate Vulnerability Assessment for the Northeast Shelf. Similar to our results, this previous score was driven by Early Life History Requirements due to the benthic egg masses of this species (Hare et al. 2016).

Exposure:

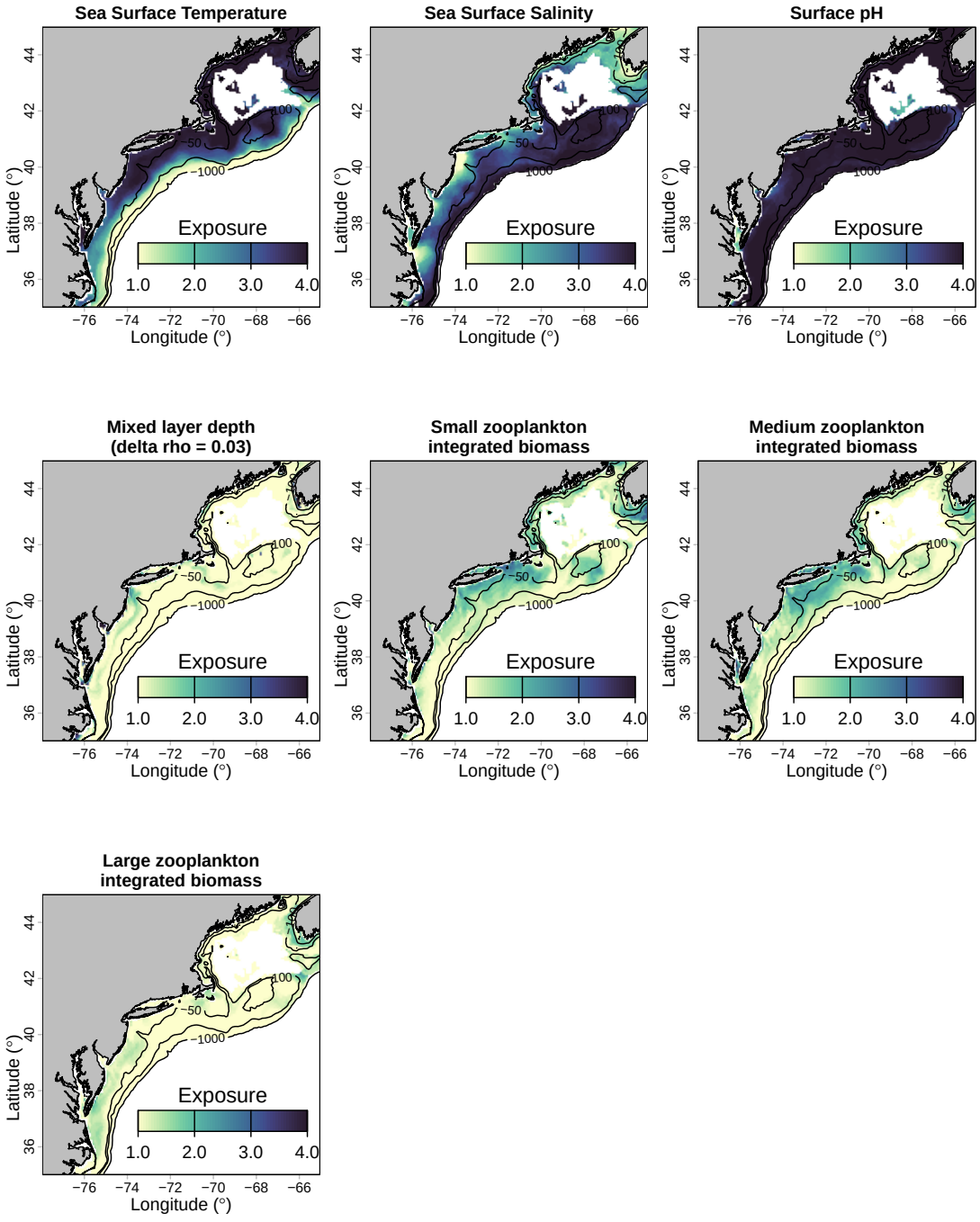


Figure 3: Maps of each Exposure Factor for Longfin squid. Exposure was calculated between 2014-2023 and 2025-2035

Mean Exposure for Longfin squid was **High** (3.02 +/- 0.54) (Figure 1). Exposure was high throughout the continental shelf. Scores were Very High over Georges Bank and Low at the mouth of Chesapeake

Bay (Figure 4). These patterns were driven by high exposure in sea surface temperatures, salinities, pH, and zooplankton biomass (Figure 3).

Longfin squid also received a High Exposure score in the previous assessment (Hare et al. 2016). This was driven by high temperature and ocean acidification scores (Hare et al. 2016).

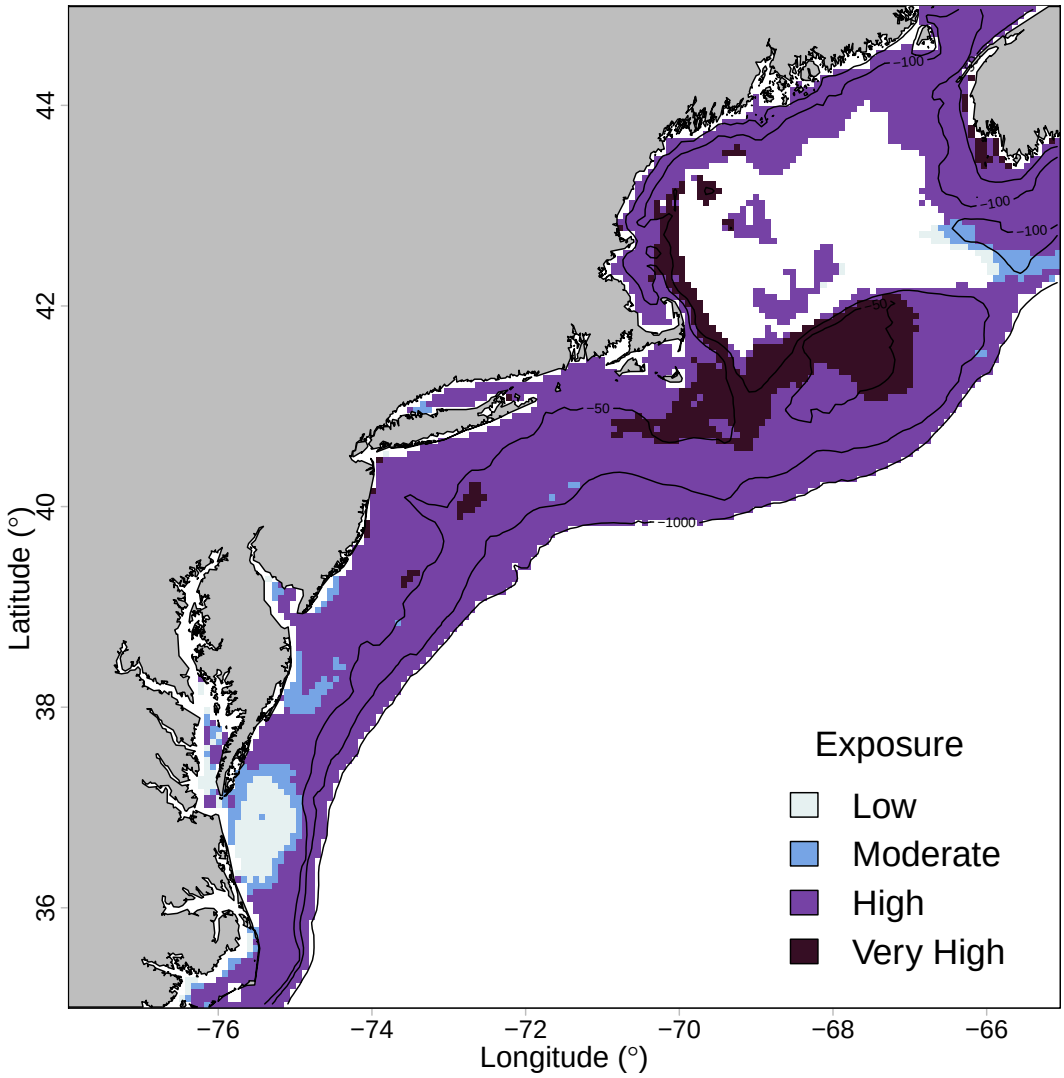


Figure 4: Total Exposure on the Northeast Shelf for Longfin squid. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

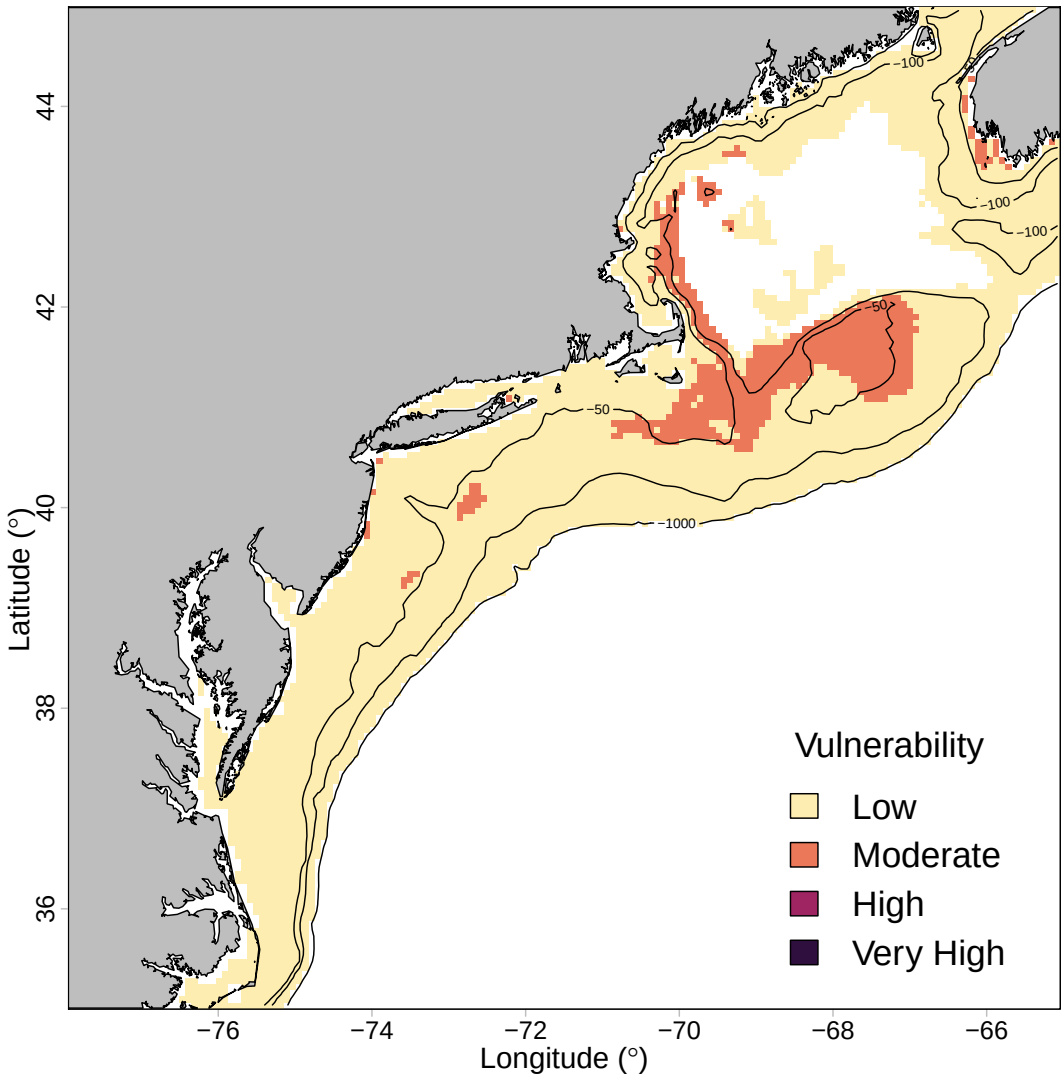


Figure 5: Vulnerability on the Northeast Shelf for Longfin squid. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean Vulnerability for Longfin squid was **Low** (3.02 +/- 0.54) (Figure 1). Vulnerability remained Low on average when Exposure was calculated only with factors that were important to its distribution (Figure 1). Vulnerability was Moderate along the west coast of the Gulf of Maine and over Georges Bank, but Low throughout the rest of the species' range (Figure 5).

Longfin squid also received a Low Vulnerability score in the previous Assessment (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

93% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified (Figure 1).

Model Confidence Score:

Mean model confidence for longfin squid was 2.2 (+/- 0.27), indicating moderate confidence in the ensemble model. Experts stated that the model captured species distributions well and environmental drivers met expectations. The model AUC score was 0.83 indicating good performance.

Additional Metrics

Distribution Shifts:

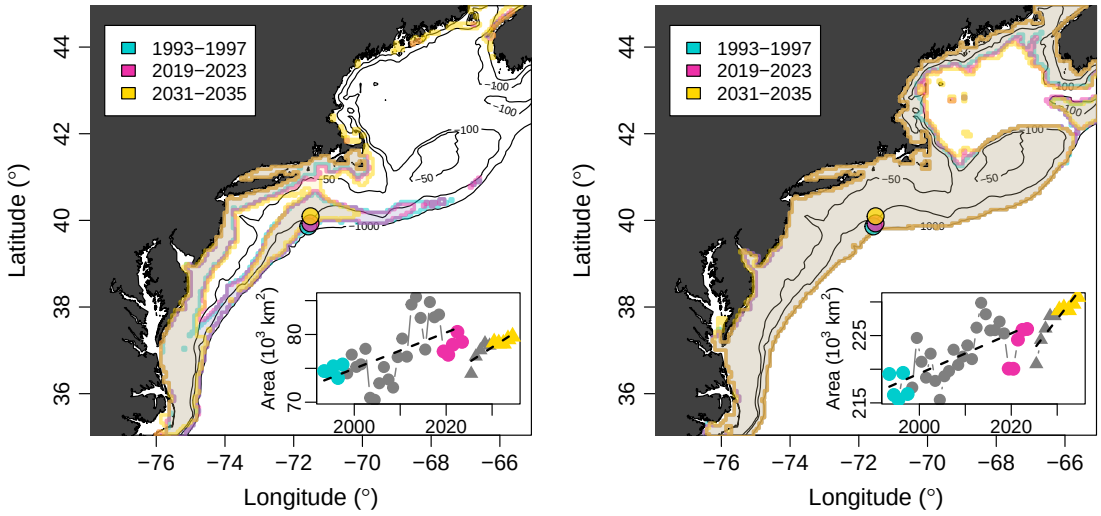


Figure 6: Distribution shifts on the Northeast Shelf for Longfin squid. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Longfin squid habitat grew significantly between 1993 and 2023. While core habitat was lost along the continental shelf, habitat area grew along the coastline (Figure 6). During this time, the weighted centroid shifted 9 km northeast.

Under near future climate conditions, habitat area is predicted to continue expanding significantly. More core habitat loss is predicted along the continental shelf but more habitat is predicted to become available

on the shelf itself and in nearshore areas (Figure 6). The weighted centroid is predicted to shift a further 18 km north.

In the previous assessment, longfin squid received a High Distributional Vulnerability Rank score due to their mobility, migrations, and habitat preferences (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	75,246	217,752	7.86	9.06
2019-2023	9.3 km NE	78,669 (+3,423)	223,390 (+5,637)	8.10 (+0.25)	9.22 (+0.16)
2031-2035	18.2 km N	79,011 (+342)	229,407 (+6,017)	8.10 (0)	9.22 (0)

Directional Effect:

Experts predicted that climate change would have **neutral** impacts on long-finned squid. Certainty was 95%. The short life-span of long-finned squid can make it challenging to understand the effects of climate change. In warmer temperatures, squid may grow faster and mature earlier, but they may also be negatively impacted by ocean acidification (Pecl and Jackson 2008) (Arkhipkin et al. 2015). In the previous assessment along the Northeast shelf, experts predicted that climate change would have a positive impact on long-finned squid, with a low amount of certainty (Hare et al. 2016).

Climate Effects on Abundance and Distribution

Cephalopods have been shown to be sensitive to environmental variability and climate change, due to their short lifespan (Moustahfid et al. 2021). Inshore squids, such as longfin, may have complex responses to climate change. As temperatures increase, squids will likely hatch smaller and earlier, grow faster, mature at a smaller size, and have shorter life spans. They may also require more food and oxygen, but a reduced capacity to cope without food (Pecl and Jackson 2008). Longfin squid recruitment may negatively be affected by impacts of increasing ocean acidification on the paralarval stage (Arkhipkin et al. 2015). Longfin squid raised under elevated pCO^2 increased hatching time (~24 hours longer) and had shorter mantle lengths as well as significantly reduced statolith surface area (by 25%) and abnormally shaped and porous statoliths (Kaplan et al. 2013). Abundance and distribution changes have been linked to the North Atlantic Oscillation, Gulf Stream, and surface fronts on the outer continental shelf (Dawe et al. 2007) (Hendrickson, Gangopadhyay, and Moustahfid 2023) (Manderson et al. 2011).

Life History Synopsis

In the Northwest Atlantic, longfin squid are most abundant from Georges Bank to Cape Hatteras although their range expands from Newfoundland to the Gulf of Venezuela (Jacobson 2005). The North Atlantic stock undertakes seasonal migrations: moving offshore in the fall to overwinter along the shelf edge and slope and migrating inshore in the spring following warming water temperatures to spawn (Hatfield and Cadrin 2002). More recent studies after the development of statolith aging techniques highlight a peak

in spawning from late spring to summer (Hatfield and Cadrin 2002) (Macy and Brodziak 2001) and hatching in late summer (Jacobson 2005). Some spawning occurs throughout the year both inshore and offshore depending on hatching month and length-at-age (Macy III 1994) (Macy and Brodziak 2001), with winter spawning thought to occur primarily south of Cape Hatteras (Hatfield and Cadrin 2002). Eggs incubate for 11 - 27 days on sandy or muddy bottoms attached to solid substrate, with faster development times are warmer temperatures. The duration of the larval stage is unclear, but they occupy coastal surface waters. Once juveniles, they transition from inhabiting surface waters to a demersal lifestyle (Jacobson 2005). The Gulf Stream and warm core rings may facilitate the recruitment transport of larval and juvenile squid northward, but potential for inputs to the population from the Southeast U.S. Continental Shelf and offshore are difficult to quantify (Hatfield and Cadrin 2002). Juveniles and adults are generally demersal and inhabit the continental shelf and upper slope to depths of 400 m. This varies seasonally, with adults occupying shallower habitats in summer and fall, and deeper habitats in the winter. Diel vertical migrations have been reported (Jacobson 2005). Juveniles consume planktonic organisms, primarily copepods. Their preferred prey type varies with size (Jacobson 2005). Larger individuals feed on crustaceans and small fish. Diets shift seasonally from crustacean-dominated in the spring to fish-dominated in the fall. Cannibalism is observed in individuals larger than 5 cm (Jacobson 2005). Longfin squid are prey for a variety of finfish, marine mammals, and birds (Jacobson 2005). Most statolith-aging studies suggest that longfin squid have a short lifespan of around 1 year or less, reaching maturity shortly before spawning and senescence (Macy and Brodziak 2001). Longfin squid are managed by the Mid-Atlantic Fishery Management Council under the Mackerel, Squid, and Butterfish Fishery Management Plan. The stock is not overfished, and overfishing reference points are not available (Mid-Atlantic Fishery Management Council (MAFMC) 2025).

Supplemental Scores

Exposure Timeseries

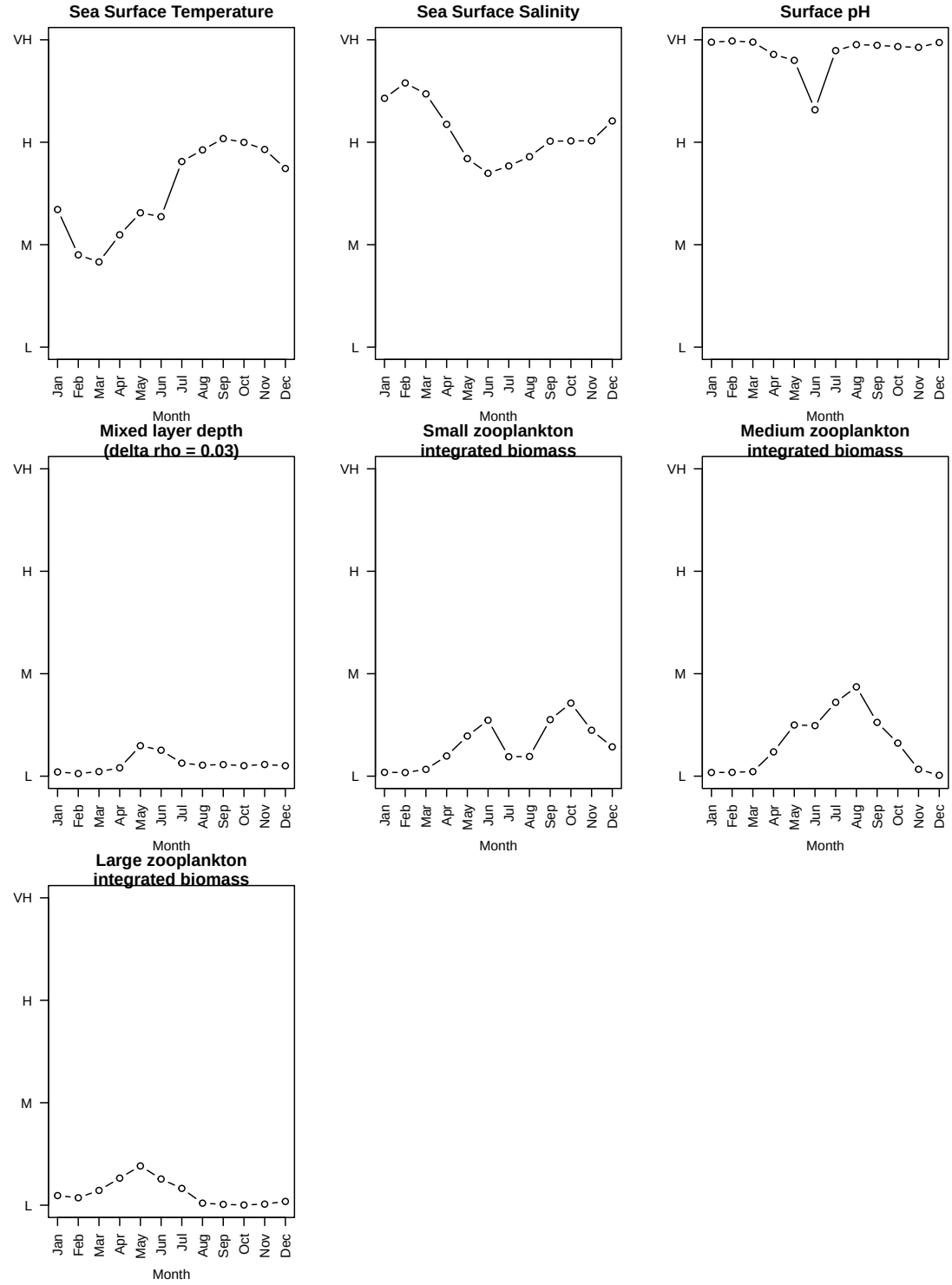


Figure 7: Monthly Timeseries of each Exposure Factor for Longfin squid. Exposure was calculated between 2014-2023 and 2025-2035

Monthly exposure ranged from Moderate to High (Figure 8). Exposure was Moderate May through August due to declines in sea surface salinity exposure (Figure 8, Figure 7).

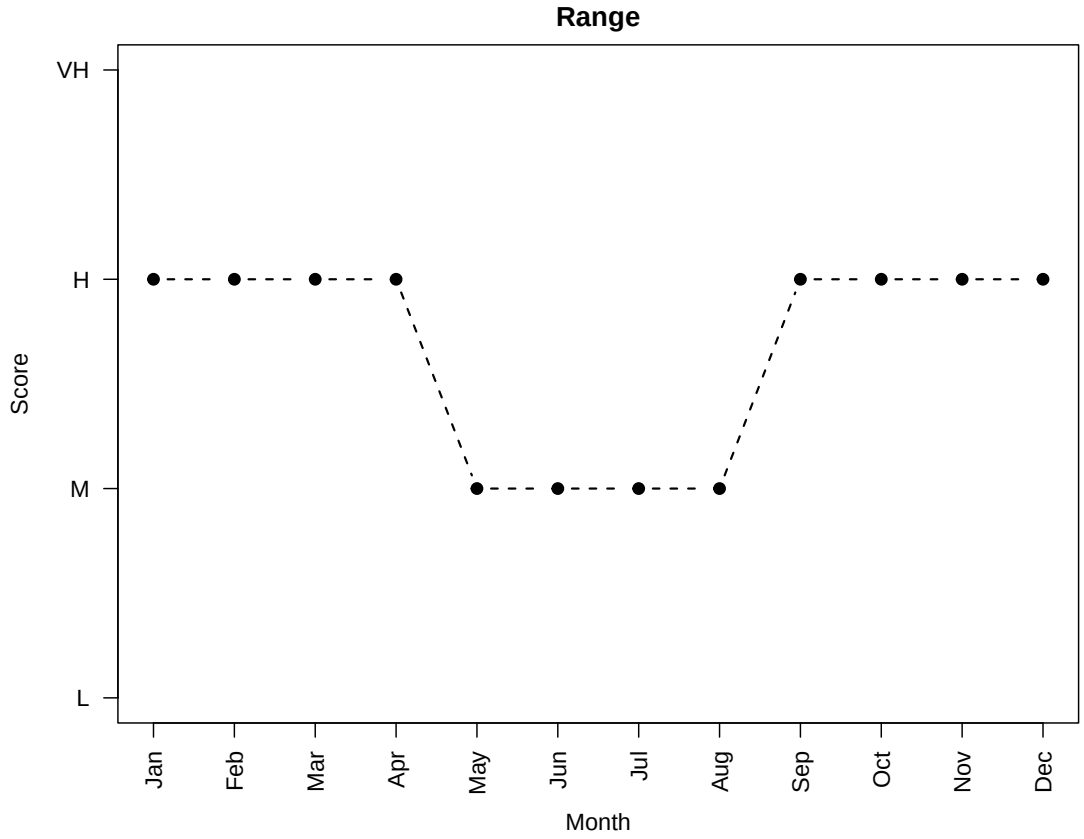


Figure 8: Timeseries of Total Exposure for Longfin squid

Exposure due to Significant Factors

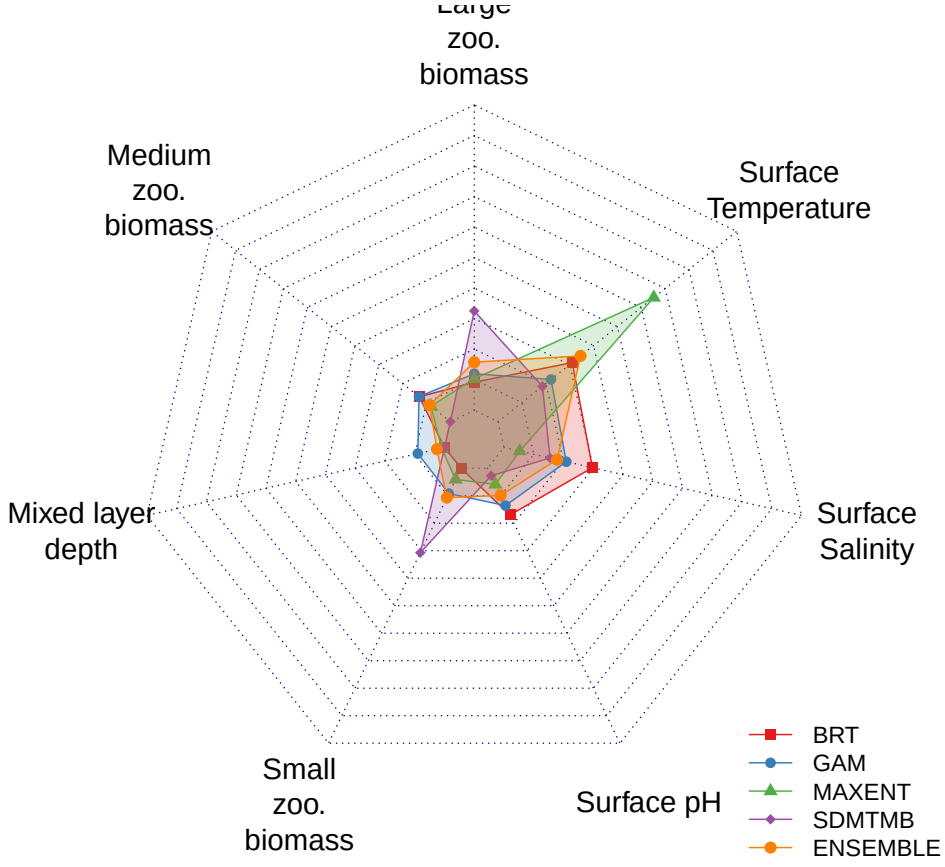


Figure 9: Dynamic Factors driving Longfin squid distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Sea surface temperature, salinity, and small and large zooplankton biomass were important to the distribution of longfin squid (Figure 9). Longfin are planktivores and prey size increases with body size (Jacobson 2005). Temperature plays an important role in development and major currents like the Gulf Stream play a large role in transport (Pecl and Jackson 2008), (Arkhipkin et al. 2015) (Hatfield and Cadrin 2002). When only these factors were used to calculate exposure, the mean score dropped to Moderate (1.7 +/- 0.87) (Figure 11). In this scenario, exposure was greatest along the middle of the continental shelf and Georges Bank, where high exposures overlapped for temperature, salinity, and zooplankton variables (Figure 10).

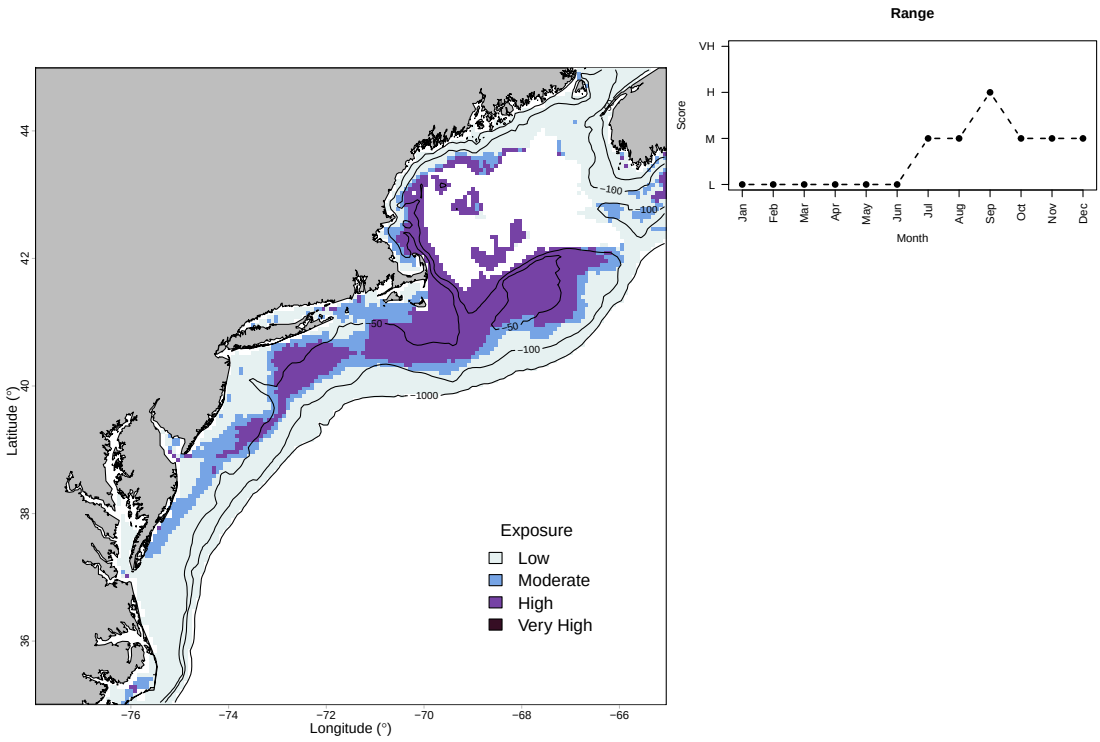


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Longfin squid distributions

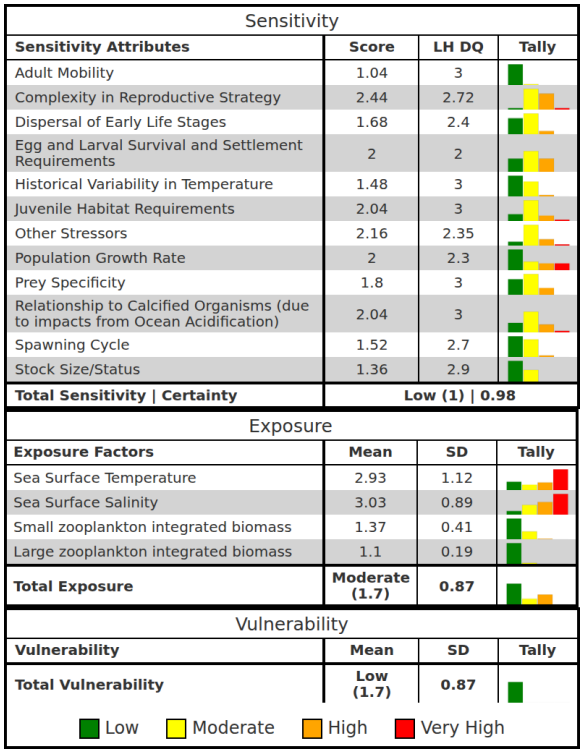


Figure 11: As in Figure 1 for Longfin squid, but with only significant Exposure factors and the resulting Vulnerability Scores.

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